ANDHRA PRADESH · SATELLITE + SENSOR FUSION

Go-Live & Handoff Checklist

Everything required to take the prototype to a hosted, auto-refreshing product with real data — and hand it off to the receiving team. Status: **demo-ready, passed external audit.**

BLOCKER for real live data DEPLOY for a clean hosted launch LATER nice-to-have / post-live

1 The only true blockers

1. **Official APWRIMS data access** — replaces the short-lived browser-session sample.
2. **Official mandal boundaries + admin IDs** — replaces the public-prototype geometry.
3. **A host + deploy hook** — or auto-deploy on push.

Everything else is configuration or polish. Architecture, pipeline, tests, and provenance are already in place.

2 Credentials & secrets

Set in the host's secret store and as GitHub Actions secrets — never in the repo (verified clean of secrets).

Secret / env var	Used by	Status	Priority
`APWRIMS_COOKIE`	`fetch_apwrims_history.py`	session cookie today; replace with official API/token	BLOCKER
`EARTHDATA_USER / _PASS`	NASA GRACE-DA fetch (`.netrc` step)	free NASA Earthdata account	DEPLOY
`DEPLOY_HOOK_URL`	weekly workflow redeploy step	host deploy webhook	DEPLOY
`ANTHROPIC_API_KEY`	ai-brief route	optional — degrades gracefully without it	LATER
`ALLOW_INSECURE_TLS`	public-data fetchers	leave **UNSET** in prod; `=1` only for local cert issues	LATER

3 Data sources & auth

Source	Provides	Auth
APWRIMS (AP-GWD)	ground-truth groundwater depth	BLOCKER official access (cookie today)
NASA GRACE-DA	groundwater storage percentile	DEPLOY Earthdata login (likely)
CHIRPS	rainfall / recharge	public
TerraClimate	ET vs rainfall (water balance)	public
NASA POWER	rainfall fallback	public API
CGWB / India-WRIS	independent validation network	public portal
Open-Elevation	terrain features	public

All non-APWRIMS sources are open; confirm each provider's rate limits and attribution/licensing terms before automated production use.

4 Replace prototype data with official data

- BLK Official **APWRIMS groundwater export/API** — swap the session-sample CSV.
- BLK Official **APWRIMS / APSAC / RTGS mandal boundaries** — replace `ap_map_geometry.json` (`boundary_source=public_prototype`).
- BLK Official **mandal admin IDs / codes** — wire into the join + records.
- DEP Once official inputs land, review and lift the "prototype / not official" caveats; set `official_flag=true` where justified.

5 Auto-refresh pipeline (built — needs hardening)

The weekly GitHub Action `phase3_weekly_levels.yml` runs Mondays: fetch → build features → predict → `build_real_app_data` → commit `app/data` + manifest → redeploy.

- BLK **APWRIMS cookie is short-lived** — until official API access exists, a human must refresh `APWRIMS_COOKIE` or the step fails.
- DEP Add the **Earthdata `.netrc` step** + secrets (placeholder already in the workflow).
- DEP **Gate publish on CI**: run `pytest` + `validate_fusion_outputs.py` BEFORE commit/deploy, so a bad fetch can't publish bad data.
- LAT **Failure alerting** (Slack/email) on a failed weekly run or stale data.

Cadence: APWRIMS readings are ~monthly, so the meaningful refresh is monthly even though the job runs weekly. The UI shows a "Data as of {month}" stamp.

```
# manual run
export APWRIMS_COOKIE='JSESSIONID=...; ...'      # authorized session only
python3 phase3_levels/fetch_weekly.py            # fetch + features + predict + publish
python3 phase3_levels/build_real_app_data.py     # (re-publish from existing outputs)
python3 -m pytest -q                             # verify before shipping
python3 scripts/validate_fusion_outputs.py
cd app && npm ci && npm run build
```

6 Hosting, security & quality gates

Hosting / deploy

- DEP Host: **Vercel / Netlify / Render** (auto-deploy on push) or self-host the **standalone** output — `node .next/standalone/server.js` (not `next start`).
- DEP Custom **domain + HTTPS**; set `DEPLOY_HOOK_URL` so the monthly commit goes live.
- LAT Optionally serve `app/data` from object storage / an API so fresh data shows without a rebuild; self-host fonts for offline builds.

Security & quality gates

- DEP Secrets only in host/CI stores (repo clean; `.gitignore` excludes caches/models/rasters).
- LAT AI endpoint rate limit is in-memory (per-process) — add a distributed limiter if public at scale; add error monitoring (Sentry).
- DEP Keep running: **51 tests** (`test_phase3_audit.py` guardrails), `validate_fusion_outputs.py` , and `dataset_manifest.json` hashes — wire as the publish gate.

7 Known limitations to communicate at handoff

- Estimates are modelled gap-fill/forecast for the **639 mandals with APWRIMS history** (temporal hold-out, ±1.3 m); ~20 mandals have no sensor series (shown grey).
- Spatial estimation of **sensorless** mandals is a research backtest, **not deployed**.
- GRACE-DA is **district-scale context** (coarse), not mandal depth. Pumping vs drought signals are **verify-first hypotheses**, not attributions.
- Confidence bands are model-derived, **not yet calibrated** for empirical p10–p90 coverage.

8 Deferred engineering & ownership runbook

Deferred (optional, post-live)

- **LAT** Consolidate the legacy V0 (`scripts/`) and Phase-3 pipelines into one path (Phase-3 is live, V0 is reference-only).
- **LAT** Automated responsive + accessibility tests (Playwright + axe); phone hamburger drawer.
- **LAT** Recalibrate on official APWRIMS data; calibrate confidence bands; validate spatial coverage if statewide sensorless estimates are wanted.

Ownership (fill in at handoff)

- [ ] Repo owner / maintainer
- [ ] Hosting account & access
- [ ] Secret store & rotation owner
- [ ] Who refreshes `APWRIMS_COOKIE` until official access
- [ ] Monthly-refresh failure escalation
- [ ] APWRIMS authorization / MoU owner (AP-WRD)

ⓘ The engineering is done and audited — the path to live is **organizational** (official APWRIMS access + boundaries) plus a **host choice**. Everything else is configuration.